

TABLE 1: DIFFERENCES BETWEEN HARDWARE PLCs, SOFTWARE PLCs, AND VIRTUAL PLCs

Features	Hardware PLCs	Software PLCs	Virtual PLCs
Hardware dependency	Require dedicated hardware	Run on industrial PLCs or embedded hardware	Run on virtual machines, cloud platforms, or edge computing devices
Performance	High	Near real-time performance	Performance depends on the network
Scalability	Limited	Less	High
Cost	High	Less	Less
Reliability	High	Good	Depends on the cloud and edge infrastructure
Real time capabilities	Excellent	Good	Depends on the cloud and edge infrastructure
Cyber security	Less exposed to cyber threats	More exposed to cyber threats	High
Flexibility	Less	More	Highly flexible
Updates	Manual	Software-based updates	Can be updated remotely

industrial PCs, edge devices, or cloud servers, using hypervisors and containerisation to create isolated environments for control applications. Real-time operating systems (RTOS) and software-based logic execution engines ensure that vPLCs maintain the deterministic performance required for industrial automation.

Fieldbus communication protocols such as EtherCAT, PROFINET, and Modbus allow seamless interaction with sensors, actuators, and other industrial devices.

A key enabler of vPLCs is industrial IoT (IIoT) and edge computing, which improve remote monitoring and predictive maintenance. Machine learning and AI-powered analytics integrated with vPLCs support production optimisation and fault detection before failures occur. Technologies like 5G and time-sensitive networking (TSN) further enhance real-time data transmission, increasing reliability in critical applications.

Cloud-based platforms such as AWS IoT Greengrass, Microsoft Azure IoT, and Siemens Industrial Edge enable scalable deployment of virtualised control systems. Together, these technologies power Industry 4.0 innovations by enabling flexible, software-defined automation solutions.

Advantages

1. Eliminate the need for dedicated PLC hardware, reducing capital and maintenance costs
2. Enable engineers to monitor, update, and control systems remotely via cloud or network access
3. Integrate easily with IoT, AI, cloud computing, and big data analytics for advanced automation
4. Can be quickly installed and configured on existing IT infrastructure, reducing setup time
5. Software updates can be performed remotely without interrupting production
6. Cloud analytics help detect and address issues before failures occur
7. Operate across various platforms, including edge devices, industrial PCs, and cloud environments
8. Reduce power consumption associated with dedicated PLC hardware
9. Allow multiple teams to access and manage vPLCs in real time across different locations

Limitations

1. Network and processing delays may affect real-time control performance
2. More vulnerable to hacking and cyber threats than traditional hardware PLCs
3. May require additional software or hardware integration

with legacy industrial systems

4. Cloud-based vPLC solutions may incur ongoing subscription costs

5. Some industries may hesitate to transition due to reliability concerns and legacy infrastructure

6. Despite virtualisation, a failure of the host hardware (server or edge device) may halt the vPLC system

7. Require skilled personnel to manage networking, virtualisation, and cloud infrastructure

Road ahead

vPLCs are expected to dominate industrial automation by 2030. These server-based controllers are initiating a paradigm shift by integrating manufacturing operations with enterprise IT systems. As industries accelerate their digital transformation journeys, vPLCs will play a pivotal role in shaping the future of automation.

Advancements in edge computing, 5G, and IIoT will further improve the performance and reliability of vPLCs. Organisations investing in software-defined automation stand to gain from increased flexibility and reduced operational costs. While traditional PLCs will remain relevant in critical applications, vPLCs will drive innovation, efficiency, and responsiveness in modern industrial environments. **EFY**